

## Chapter 30 Check

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## 30 Supergraphs

The introduction of Feynman graphs in the late 1940s provided the enormous advantage of preserving manifest Lorentz invariance at every stage of perturbative calculations. For this purpose it was necessary that the exchange of all spin states of any virtual particle should be described by a single propagator. Fortunately, it has become possible to go a step further, and develop a supergraph formalism, in which supersymmetry as well as Lorentz invariance is kept manifest at every stage [1]. To do so, it is necessary to describe the exchange of all the particles described by a given superfield by a single superpropagator.

There is a problem here in that the left-chiral superfields  $\Phi$  over which we integrate are subject to a differential constraint,  $\mathcal{D}_R\Phi = 0$ . This is analogous to the problem in electrodynamics that the field-strength tensor is subject to a differential constraint, the homogeneous Maxwell equation. In electrodynamics this problem is dealt with by expressing the field-strength tensor in terms of a vector potential, and doing path integrals over the vector potential instead of the field-strength tensor. In much the same way, here we will impose the left-chiral constraint by expressing left-chiral superfields in terms of potential superfields, over which we integrate in path integrals. In this formalism we encounter problems like those produced by gauge invariance in electrodynamics, and deal with them in much the same way.

The most important result to come out of the supergraph formalism was the non-renormalization theorem for the superpotential [2]. We have already proved this theorem in Section 27.6, using a much easier indirect technique developed by Seiberg, which can be extended also to describe non-perturbative effects. Nevertheless, it is interesting to see how these surprising cancellations of renormalization effects occur in actual perturbative calculations.

### 30.1 Potential Superfields

Consider a theory of left-chiral superfields  $\Phi_n(x, \theta)$  and their complex conjugates, but for simplicity without gauge superfields. All of the vacuum expectation values of time-ordered products of component fields may be calculated from vacuum expectation values of the

time-ordered products of these superfields. We might try to calculate these from the path-integral formula

$$\begin{aligned} & \langle T \{ \Phi_{n_1}(x_1, \theta_1), \Phi_{n_2}(x_2, \theta_2), \dots \} \rangle \\ &= \int \left[ \prod_{n,x,\theta} d\Phi_n(x, \theta) \right] \exp(iI[\Phi]) \Phi_{n_1}(x_1, \theta_1) \Phi_{n_2}(x_2, \theta_2) \cdots \end{aligned} \quad (30.1.1)$$

where  $I[\Phi]$  is the action

$$I[\Phi] = \frac{1}{2} \int d^4x \left[ \sum_n \Phi_n^*(x, \theta) \Phi_n(x, \theta) \right]_D + 2 \operatorname{Re} \int d^4x [f(\Phi)]_{\mathcal{F}}. \quad (30.1.2)$$

(The factor  $1/2$  is introduced in the first term, as in Eq. (26.4.3), to give the component fields of  $\Phi$  a conventional normalization.) But we cannot simply read off the supergraph Feynman rules from Eq. (30.1.1), because the functional integral over the superfields  $\Phi_n$  must be constrained to satisfy the left-chiral condition  $\mathcal{D}_R \Phi_n = 0$ .

This is analogous to a similar problem in electrodynamics. As discussed in Section 12.3, at energies below the electron mass the interactions of soft photons with each other are described by an effective action of the form:

$$I[f] = -\frac{1}{4} \int d^4x \left[ f_{\mu\nu} f^{\mu\nu} + c_1 (f_{\mu\nu} f^{\mu\nu})^2 + c_2 (\epsilon_{\mu\nu\rho\sigma} f^{\mu\nu} f^{\rho\sigma})^2 \right].$$

But we cannot read off the Feynman rules from this formula without taking into account the fact that the path integral is constrained by the homogeneous Maxwell equations

$$\partial_\mu f_{\nu\rho} + \partial_\nu f_{\rho\mu} + \partial_\rho f_{\mu\nu} = 0.$$

As everyone knows, we deal with this constraint by introducing a four-vector potential  $A_\mu$ , with  $f_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ , so that the constraint is automatically satisfied, and integrate over  $A_\mu(x)$ , not  $f_{\mu\nu}(x)$ .

In the same way, we may adopt a trick already used in Section 26.6 to derive the field equations for superfields, and introduce a non-chiral potential superfield  $S_n(x, \theta)$ , with

$$\Phi_n = \mathcal{D}_R^2 S_n, \quad (30.1.3)$$

where

$$\mathcal{D}_R^2 \equiv \sum_{\alpha\beta} \epsilon_{\alpha\beta} \mathcal{D}_{R\alpha} \mathcal{D}_{R\beta}, \quad (30.1.4)$$

so that  $\Phi_n$  automatically satisfies the left-chiral constraint,  $\mathcal{D}_{R\alpha} \Phi_n = 0$ . In place of Eq. (30.1.1), we then have the path-integral formula

$$\begin{aligned} & \langle T \{ \Phi_{n_1}(x_1, \theta_1), \Phi_{n_2}(x_2, \theta_2), \dots \} \rangle \\ &= \int \left[ \prod_{n,x,\theta} dS_n(x, \theta) \right] \exp(iI[\mathcal{D}_R^2 S]) \mathcal{D}_R^2 S_{n_1}(x_1, \theta_1) \mathcal{D}_R^2 S_{n_2}(x_2, \theta_2) \cdots \end{aligned} \quad (30.1.5)$$

In expressing the action (30.1.2) in terms of  $S_n$ , we may recall that the  $D$ -term of a superderivative does not contribute to the action, so we may shift the operator  $(\mathcal{D}_R^2)^* = \mathcal{D}_L^2$  acting on  $S_n^*$  in the first term of Eq. (30.1.2) to act instead on  $S_n$ , which gives

$$I[\mathcal{D}_R^2 S] = \frac{1}{2} \int d^4x \left[ \sum_n S_n^* \mathcal{D}_L^2 \mathcal{D}_R^2 S_n \right]_D + 2 \operatorname{Re} \int d^4x [f(\mathcal{D}_R^2 S)]_{\mathcal{F}}. \quad (30.1.6)$$

The  $\mathcal{D}_R^2$  operator in any one of the factors  $\mathcal{D}_R^2 S$  in any term in  $f(\mathcal{D}_R^2 S)$  may be taken to act on the whole term, since  $\mathcal{D}_R$  gives zero when acting on any other factor of  $\mathcal{D}_R^2 S$  in that term. In this way, we may write

$$f(\mathcal{D}_R^2 S) = \mathcal{D}_R^2 \tilde{f}(S), \quad (30.1.7)$$

where  $\tilde{f}(S)$  is obtained from  $f(\mathcal{D}_R^2 S)$  by omitting any one operator  $\mathcal{D}_R^2$  in each term. For instance, for a single type of superfield, if  $f(\Phi) = \sum_r c_r \Phi^r$ , then

$$\tilde{f}(S) = \sum_r c_r S (\mathcal{D}_R^2 S)^{r-1}.$$

Using Eqs. (30.1.6), (30.1.7), and (26.3.31), we may put the whole action in the form of a  $D$ -term

$$I[\mathcal{D}_R^2 S] = \frac{1}{2} \int d^4x \sum_n [S_n^* \mathcal{D}_L^2 \mathcal{D}_R^2 S_n]_D + 2 \operatorname{Re} \int d^4x [\tilde{f}(S)]_D. \quad (30.1.8)$$

Eq. (26.6.5) shows that this may also be written as a superspace integral:

$$I[\mathcal{D}_R^2 S] = -\frac{1}{4} \int d^4x \int d^4\theta \sum_n S_n^* \mathcal{D}_L^2 \mathcal{D}_R^2 S_n - \operatorname{Re} \int d^4x \int d^4\theta \tilde{f}(S). \quad (30.1.9)$$

This is the action we will use in the path-integral derivation of the supergraph formalism.

### 30.2 Superpropagators

Usually the propagator can be obtained directly from the part of the action of second order in the fields. If we write this quadratic part for a complex field  $\phi_i$  (where  $i$  is a compound index including spacetime coordinates as well as spin and species labels) in the form

$$I_{\text{quad}}[\phi] = - \sum_{ij} D_{ij} \phi_i^* \phi_j, \quad (30.2.1)$$

with  $D_{ij}$  Hermitian, then as explained in Section 9.4 the propagator is simply  $\Delta = D^{-1}$ . The problem with this arises when the quadratic part of the action is invariant under a (linearized) gauge transformation

$$\phi_i \longrightarrow \phi_i + \xi_i, \quad (30.2.2)$$

for some class of ‘vectors’  $\xi$ . In this case we have

$$\sum_i D_{ij} \xi_j = 0, \quad (30.2.3)$$

and we clearly cannot invert the ‘matrix’  $D_{ij}$ . In electrodynamics the problem arises because the Lagrangian density is invariant under a gauge transformation  $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ . We have this problem here too: since the action is actually a functional of  $\mathcal{D}_R^2 S_n$  and not of  $S_n$  itself, it is invariant under the transformation

$$S_n \rightarrow S_n + \mathcal{D}_R X_n, \quad (30.2.4)$$

for any superfields  $X_n$ .

In the electrodynamics of charged particles the problems introduced by gauge invariance are typically handled by choosing a gauge, for instance by the Faddeev–Popov–de Witt method described in Section 15.5. But the problems raised here by invariance under the transformation (30.2.2) are much more like the problems of the effective field theory of photons at energies below the threshold for producing charged particles, where the theory is gauge-invariant simply because the action only involves gauge-invariant fields. In such theories there is a simpler option. In addition to the eigenvector  $\xi_i$  of  $D_{ij}$  with eigenvalue zero (taken for simplicity to be in a unique direction) we can find a set of orthonormal eigenvectors  $u_{\nu i}$  with eigenvalues  $d_\nu \neq 0$ :

$$\sum_j D_{ij} u_{\nu j} = d_\nu u_{\nu i}, \quad \sum_i u_{\nu i}^* u_{\nu' i} = \delta_{\nu \nu'}, \quad \sum_i u_{\nu i}^* \xi_i = 0. \quad (30.2.5)$$

We can introduce a new set of integration variables  $\phi'$  and  $\phi'_\nu$ :

$$\phi_i = \phi' \xi_i + \sum_\nu \phi'_\nu u_{\nu i}. \quad (30.2.6)$$

The sort of integral encountered in Feynman diagram calculations of quantum expectation values may then be written

$$\begin{aligned} & \int \left[ \prod_i d\phi_i d\phi_i^* \right] \exp\{iI_{\text{quad}}[\phi]\} \phi_a \cdots \phi_b^* \cdots \\ &= \mathcal{J} \int d\phi' d\phi'^* \int \left[ \prod_\nu d\phi'_\nu d\phi'^*_\nu \right] \exp \left\{ -i \sum_\nu d_\nu |\phi'_\nu|^2 \right\} \\ & \quad \cdot \left[ \phi' \xi_a + \sum_\nu \phi'_\nu u_{\nu a} \right] \cdots \left[ \phi' \xi_b + \sum_\nu \phi'_\nu u_{\nu b} \right]^* \cdots, \end{aligned} \quad (30.2.7)$$

where  $\mathcal{J}$  is the Jacobian of the transformation (30.2.6). The integral over  $\phi'$  and  $\phi'^*$  is of course ill defined, because they do not appear in the argument of the exponential. But this doesn’t matter if the action only involves gauge-invariant fields, because then  $\phi_a, \phi_b$ , etc. will be contracted with ‘currents’  $J_a, J_b$ , etc. for which

$$\sum_a \xi_a J_a = 0. \quad (30.2.8)$$

We may thus write Eq. (30.2.7) as

$$\begin{aligned}
& \int \left[ \prod_i d\phi_i d\phi_i^* \right] \exp\{iI_{\text{quad}}[\phi]\} \phi_a \cdots \phi_b^* \cdots \\
&= \mathcal{C} \int \left[ \prod_\nu d\phi'_\nu d\phi'^*_\nu \right] \exp\left\{-i \sum_\nu d_\nu |\phi'_\nu|^2\right\} \left[ \sum_\nu \phi'_\nu u_{\nu a} \right] \cdots \left[ \sum_\nu \phi'_\nu u_{\nu b} \right]^* \cdots \\
&\quad + \xi\text{-terms},
\end{aligned} \tag{30.2.9}$$

where ‘ $\xi$ -terms’ denotes terms proportional to one or more factors of  $\xi_a, \xi_b$ , etc., which vanish when contracted with  $J$ s satisfying Eq. (30.2.8), and  $\mathcal{C}$  is the infinite constant  $\mathcal{J} \int d\phi'$ . The integral over the  $\phi'_\nu$  then gives

$$\begin{aligned}
& \int \left[ \prod_i d\phi_i d\phi_i^* \right] \exp\{iI_{\text{quad}}[\phi]\} \phi_a \cdots \phi_b^* \cdots \\
&\propto \sum_{\text{pairings}} [-i\Delta_{ab}] \cdots + \xi\text{-terms},
\end{aligned} \tag{30.2.10}$$

where the sum over pairings is over all ways of pairing indices on the  $\phi$ s with indices on the  $\phi^*$ s, and  $\Delta_{ab}$  is the propagator

$$\Delta_{ab} = \sum_\nu \frac{u_{\nu a} u_{\nu b}^*}{d_\nu}. \tag{30.2.11}$$

Instead of actually evaluating the sum (30.2.11), we can use its defining property that

$$\sum_c D_{ac} \Delta_{cb} = \sum_\nu u_{\nu a} u_{\nu b}^* \equiv \Pi_{ab}, \tag{30.2.12}$$

with  $\Pi$  the projection operator on the space orthogonal to  $\xi$ :

$$\Pi^2 = \Pi, \quad \Pi\xi = 0. \tag{30.2.13}$$

The solution of Eq. (30.2.12) is unique only up to  $\xi$ -terms, but these do not matter if the fields  $\phi_i$  appear in the action only in gauge-invariant combinations.

For instance, in electrodynamics, we can write the kinematic part of the action as

$$I_{\text{quad}}[A] = -\frac{1}{4} \int d^4x f_{\mu\nu} f^{\mu\nu} = +\frac{1}{2} \int d^4x A^\mu (\Box \delta_\mu^\nu - \partial_\mu \partial^\nu) A_\nu.$$

The differential operator  $-\Box \delta_\mu^\nu + \partial^\nu \partial_\mu$  is not invertible, because it has a zero eigenvalue with eigenvectors of the form  $\xi^\mu = \partial^\mu \Lambda$ . The projection matrix onto the space orthogonal to these vectors is

$$\Pi_\mu{}^\nu(x, y) = [\delta_\mu^\nu - \partial_\mu \partial^\nu \Box^{-1}] \delta^4(x - y),$$

where  $\Box^{-1} \delta^4(x - y)$  is any solution of the equation  $\Box[\Box^{-1} \delta^4(x - y)] = \delta^4(x - y)$ . The defining equation for the propagator is

$$(-\Box \delta_\mu^\lambda + \partial_\mu \partial^\lambda) \Delta_\lambda{}^\nu(x, y) = \Pi_\mu{}^\nu(x, y),$$

with the solution

$$\Delta_\lambda^\nu(x, y) = \delta_\mu^\nu \Delta_F(x - y) + \partial_\mu \partial^\nu \text{-terms},$$

where  $\Delta_F(x - y)$  is the usual Feynman propagator (6.2.16), satisfying  $\square \Delta_F(x - y) = -\delta^4(x - y)$ . (The factor 1/2 in the action does not appear in the defining equation for the propagator here because  $A_\mu$  is a real field. The origin of the  $-i\epsilon$  in the denominator of the Fourier integral in Eq. (6.2.16) is explained in the path-integral formalism in Section 9.2.)

Inspection of the first term in Eq. (30.1.9) shows that the defining equation for the superpropagator of the potential superfields is

$$\frac{1}{4} \mathcal{D}_L^2 \mathcal{D}_R^2 \Delta_{nm}^S(x, \theta; x', \theta') = \mathcal{P} \delta^4(x - x') \delta^4(\theta - \theta') \delta_{nm}, \quad (30.2.14)$$

where  $\mathcal{P}$  is a superspace differential operator satisfying the conditions for a projection operator

$$\mathcal{P}^2 = \mathcal{P}, \quad \mathcal{P} \mathcal{D}_R = 0, \quad (30.2.15)$$

and  $\delta^4(\theta - \theta')$  is the fermionic delta function introduced in Eq. (26.6.8). The solution is

$$\mathcal{P} = -\frac{1}{16\square} \mathcal{D}_L^2 \mathcal{D}_R^2. \quad (30.2.16)$$

(It is obvious that  $\mathcal{P} \mathcal{D}_R = 0$ . To check that  $\mathcal{P}^2 = \mathcal{P}$ , we need to use Eq. (26.6.12), which shows that  $\mathcal{D}_R^2 \mathcal{D}_L^2 \mathcal{D}_R^2 = -16\square \mathcal{D}_R^2$ .) The solution of Eq. (30.2.14) is

$$\begin{aligned} \Delta_{nm}^S(x, \theta; x', \theta') &= -\frac{1}{4\square} \delta^4(x - x') \delta^4(\theta - \theta') \delta_{nm} \\ &= \frac{1}{4} \Delta_F(x - x') \delta^4(\theta - \theta') \delta_{nm} + \mathcal{D}_R \text{-terms}. \end{aligned} \quad (30.2.17)$$

This is the superpropagator for a line created by a potential superfield  $S_m^*(x', \theta')$  and destroyed by a potential superfield  $S_n(x, \theta)$ , which we must use in evaluating supergraphs for the action (30.1.9). To make contact with ordinary propagators, it is also of some interest to consider the superpropagator of the line created by a left-chiral superfield  $\Phi_m^*(x', \theta')$  and destroyed by a left-chiral superfield  $\Phi_n(x, \theta)$ . These chiral superfields are obtained by operating on  $S_n(x, \theta)$  with  $\mathcal{D}_R^2$  and on  $S_m^*(x', \theta')$  with  $(\mathcal{D}_R^2)^* = \mathcal{D}_L'^2$ , so the propagator for the left-chiral superfields is

$$\Delta_{nm}^\Phi(x, \theta; x', \theta') = \frac{1}{4} \mathcal{D}_R^2 \mathcal{D}_L'^2 \Delta_F(x - x') \delta^4(\theta - \theta') \delta_{nm}. \quad (30.2.18)$$

For instance, the term in the superpropagator of zeroth order in  $\theta$  and  $\theta'$  is

$$[\Delta_{nm}^\Phi(x, \theta; x', \theta')] |_{\theta=\theta'=0} = \frac{1}{4} \left( \frac{\partial}{\partial \theta_R} \right)^2 \left( \frac{\partial}{\partial \theta_L'} \right)^2 \Delta_F(x - x') \delta^4(\theta - \theta') \delta_{nm}. \quad (30.2.19)$$

To evaluate this, we recall Eq. (26.6.8) for the fermionic delta function:

$$\delta^4(\theta - \theta') = \frac{1}{4} ((\theta_L - \theta_L')^T \epsilon (\theta_L - \theta_L')) ((\theta_R - \theta_R')^T \epsilon (\theta_R - \theta_R')),$$

from which we find  $(\partial/\partial \theta_R)^2 (\partial/\partial \theta_L')^2 \delta^4(\theta - \theta') = 4$ . Eq. (30.2.19) therefore yields

$$[\Delta_{nm}^\Phi(x, \theta; x', \theta')] |_{\theta=\theta'=0} = \Delta_F(x - x') \delta_{nm}, \quad (30.2.20)$$

which is just the usual propagator for the scalar components of the superfield.

### 30.3 Calculations with Supergraphs

We now consider how to use the results of the previous sections to calculate the quantum effective action  $\Gamma[S, S^*]$  for a set of classical potential superfields  $S_n(x, \theta)$  and their adjoints. Following the prescription discussed in Section 16.1, this may be defined in terms of a sum over all connected one-particle-irreducible supergraphs, consisting of vertices to which are attached directed internal and external lines. For each external line of type  $n$  ending or beginning in a vertex labelled  $x, \theta$ , we include a c-number factor  $S_n(x, \theta)$  or  $S_n^*(x, \theta)$ , respectively (but no propagator). A vertex labelled  $x, \theta$  with  $N$  incoming or  $N$  outgoing lines labelled  $n_1, n_2, \dots, n_N$  yields a factor equal to  $i$  times the coefficient of the  $S_1 S_2 \cdots S_N$  term in the superpotential  $\tilde{f}(S)$  or times the complex conjugate of this coefficient, respectively. An internal line of any type coming out of a vertex labelled  $x, \theta$  and going into a vertex labelled  $x', \theta'$  yields a propagator factor, given by Eqs. (30.2.10) and (30.2.17) as

$$-\frac{i}{4}\delta^4(\theta - \theta')\Delta_F(x - x'). \quad (30.3.1)$$

In addition, as shown by Eq. (30.1.7), a superderivative  $\mathcal{D}_R^2$  acts on the propagators or external line  $S$ -factors of all but one of the internal or external lines coming into any vertex, and a superderivative  $\mathcal{D}_L^2$  acts on the propagators or external-line  $S^*$ -factors of all but one of the lines coming out of any vertex. The product of these factors must be integrated over all  $x$ s and  $\theta$ s; the quantum effective action is the sum of these integrals for all one-particle-irreducible graphs.

By integrating by parts in superspace, the  $\mathcal{D}_L^2$  and/or  $\mathcal{D}_R^2$  operators accompanying any one propagator (say, one that connects vertices labelled  $x, \theta$  and  $x', \theta'$ ) may be moved to other propagators or external line factors. This leaves the factor contributed by this internal line proportional to a factor  $\delta^4(\theta - \theta')$ . Integrating over  $\theta'$  then eliminates this delta function, and lets us replace  $\theta'$  with  $\theta$  everywhere else. (In cases where several internal lines connect the same pair of vertices, we need to use the property of fermionic delta functions that  $[\delta^4(\theta - \theta')]^2 = 0$ .) Continuing in this way, we wind up with a single four-dimensional  $\theta$  integral, with all  $\mathcal{D}$ s acting on the external line factors  $S_n$  and  $S_n^*$ . That is, though not usually local in spatial coordinates,  $\Gamma[S, S^*]$  is *local in the fermionic coordinates*.

The structure of this functional is governed by its invariance under the ‘gauge’ transformations (30.2.4). This tells us that every one of the external line factors  $S_n$  or  $S_n^*$  must be acted on by an operator  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$ , respectively, with two possible exceptions. The exceptions are that a term with only incoming or only outgoing external lines, in which all but one of the external line factors  $S_n$  or  $S_n^*$  are acted on by  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$ , respectively, and by no other superderivatives, is still invariant under the transformation (30.2.4), even though it cannot be expressed as a four-dimensional  $\theta$ -integral of a functional only of the  $\Phi_n$  and/or  $\Phi_n^*$  alone. This is because the change in such an amplitude under this transformation could only come from the change in the external line factor  $S_n$  or  $S_n^*$  that is not acted on by an operator  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$ , and this change is eliminated if we use integration by parts to move one of the other  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$  operators to act on it. As we saw in Section 30.1, such a term in  $\Gamma[S, S^*]$  is the  $\mathcal{F}$ -term of a functional of the  $\Phi_n$  or of the  $\Phi_n^*$ , and thus makes a correction to the superpotential or to its complex conjugate. Aside from these exceptions, every term



in  $\Gamma[S, S^*]$  may be written as a four-dimensional  $\theta$ -integral of a functional only of the  $\Phi_n$  and/or  $\Phi_n^*$ , and therefore makes a correction to the  $D$ -term part of the effective action.

Also note that a term in the quantum effective action in which all but one of the external line factors  $S_n$  or  $S_n^*$  is acted on by  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$ , respectively, and also by additional  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$  operators (such as in combinations like  $\mathcal{D}_R^2 \mathcal{D}_L^2 \mathcal{D}_R^2 S_n$  or  $\mathcal{D}_L^2 \mathcal{D}_R^2 \mathcal{D}_L^2 S_n^*$ ) may also be written by integration by parts as a term in which one of the extra  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$  acts on the previously undifferentiated external line  $S_n$  or  $S_n^*$  factor. Such a term may therefore be expressed as a four-dimensional  $\theta$ -integral of a functional only of the  $\Phi_n$  and/or  $\Phi_n^*$ , and therefore just makes another correction to the  $D$ -term part of the action. The only terms in  $\Gamma[S, S^*]$  that cannot be written in this way are those that have  $E$  incoming external lines and only  $E - 1$  operators  $\mathcal{D}_R^2$  operating on the  $S_n$  external line factors, or  $E^*$  incoming external lines and only  $E^* - 1$  operators  $\mathcal{D}_L^2$  operating on the  $S_n^*$  external line factors. We can therefore tell whether a supergraph can make a contribution to the superpotential or its adjoint by simply counting the number of  $\mathcal{D}_R^2$  or  $\mathcal{D}_L^2$  operators contributed by the supergraph to the corresponding term in  $\Gamma[S, S^*]$ .

Let us count these superderivatives. Consider a connected graph with:  $V_n$  vertices with  $n$  lines coming in;  $V_n^*$  vertices with  $n$  lines going out;  $I$  internal lines;  $E$  external incoming lines; and  $E^*$  external outgoing lines. These numbers are related by

$$I + E = \sum_n n V_n, \quad I + E^* = \sum_n n V_n^*. \quad (30.3.2)$$

The total numbers of  $\mathcal{D}_R^2$  and  $\mathcal{D}_L^2$  operators are then

$$N_R = \sum_n V_n (n - 1) = I + E - V = L + V^* + E - 1, \quad (30.3.3)$$

and

$$N_L = \sum_n V_n^* (n - 1) = I + E^* - V^* = L + V + E^* - 1, \quad (30.3.4)$$

where  $V = \sum_n V_n$  is the total number of vertices with incoming lines;  $V^* = \sum_n V_n^*$  is the total number of vertices with outgoing lines; and  $L = I - V - V^* + 1$  is the number of loops. We see that a graph with any loops will have  $N_R \geq E$  and  $N_L \geq E^*$ , so that there are at least enough  $\mathcal{D}_R^2$  operators to convert all  $S_n$  into  $\Phi_n = \mathcal{D}_R^2 S_n$  and its derivatives, and enough  $\mathcal{D}_L^2$  operators to convert all  $S_n^*$  into  $\Phi_n^* = \mathcal{D}_L^2 S_n^*$  and its derivatives. *Any graph with loops thus yields only a contribution to the four-dimensional  $\theta$ -integral—in other words the  $D$ -term—of a functional of the left-chiral superfields  $\Phi_n$  and their adjoints.*

The only way to obtain a contribution to an  $\mathcal{F}$ -term or its adjoint is to have just  $N_R = E - 1$  operators  $\mathcal{D}_R^2$  or just  $N_L = E^* - 1$  operators  $\mathcal{D}_L^2$ , respectively. According to Eqs. (30.3.3) and (30.3.4), such a graph would have  $L = 0$  and  $V^* = 0$  or  $V = 0$ , respectively. In other words, since we are considering only one-particle-irreducible graphs, we can only get an  $\mathcal{F}$ -term from a graph with a single vertex with incoming lines, and we can only get its adjoint from a graph with a single vertex with outgoing lines. This contribution is nothing but the integrated  $\mathcal{F}$ -term of the original superpotential, or its adjoint. Thus we see again that *there is no finite or infinite renormalization of  $\mathcal{F}$ -terms to any order of perturbation theory.*

## Problems

1. Use Eq. (30.2.18) to calculate the propagators of the spinor and auxiliary components of a chiral superfield.
2. Consider a supersymmetric theory of a single left-chiral superfield  $\Phi$ , with Lagrangian density

$$\mathcal{L} = \frac{1}{2} [\Phi^* \Phi]_D + 2 \operatorname{Re}(g[\Phi^3]_{\mathcal{F}}),$$

with  $g$  an arbitrary complex constant. Use the supergraph formalism to calculate the one-loop contribution to the quantum effective action. Express the answer as an integral over coordinates and over a single Grassman coordinate  $\theta$ .

3. What is the superpropagator for a gauge superfield  $V(x, \theta)$  in a supersymmetric Abelian gauge theory with kinematic term (27.3.17)?

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